Developing Applications-Google

- You want to get an app on the cloud, the **Google App Engine** is the perfect tool to use to make this dream become reality.
- In essence, you write a bit of code in **Python, tweak some HTML code**, and then you've got your app built, and it only takes a few minutes.
- Having knowledge of Python certainly helps, but it isn't a deal-breaker, because Python is a lot like other scripting languages.
- **Java is very prevalent on the cloud**. It is a very robust scripting tool and one that programmers know well. But its complexity is probably hurting it more than helping.
- On average, hosting for Java applications begins around US$10 per month, while Python services start at around US$2 per month.
- Google was able to tweak the language slightly so that dangerous operations are not allowed, like writing to the file system. This prevents robust uploading services and spawn subthreads.

Developing Applications-Google

- Google built this in by design, given the fact that new programmers are likely to screw up and make an app that falls into an endless loop. **App Engine is better for the front ends of databases** that don't have to do a lot of independent thinking or computation.
- The database is integrated well with Python, but only allows basic search and store functions that you would need to tuck away users' information.
- Data objects are set up in Python, and then you use the save method and all the data disappears into the cloud where instances of the app can find it.
- Python looks a lot like SQL, but with a different syntax. That means that you can't use any of the millions of already scripted SQL tools to generate reports or produce graphs.
- **App Engine doesn't store joins**, which will break some of the code written for traditional databases. Google App Engine isn't perfect.
- The documentation mentions **web services and Asynchronous JavaScript and XML (AJAX)**, but there isn't much support for them.

Developing Applications-Google

Payment :

- Google is charging when applications exceed certain limits. For instance, Google says that you can only get "200 million megacycles of CPU per day."
- This can be a little frightening, because Google can skew the number in odd ways that are beyond your control. Further, you could see resources being used up as the database stores information on more than one server.
- One server starts asking for information, interserver traffic can slow everything down, and if two users are trying to get at the same data at the same time, access can become slow and expensive. On the plus side, App Engine will bring up new servers when demand rises.
- Google has the power to do whatever they want with your creation. Lock-in can be a problem, but at least it is somewhat offset by the open-source nature of the scripting language.
- **App Engine is best for simple applications that plan on staying simple**. The cloud can scale the application as needed, but if you have dreams of making your application big, Google says it reserves the right to "pre-screen, review, flag, filter, modify, refuse or remove any or all Content from the Service."

Developing Applications-Google

Force.com and Google :

- **Salesforce.com struck up a strategic alliance** with Google with the availability of Force.com for Google App Engine. Force.com for Google App Engine is a set of tools and services to enable developer success with application development in the cloud.

- The offering brings together Force.com and Google App Engine, enabling the creation of entirely **new web and business applications.**

- Force.com for Google App Engine builds on the relationship between Salesforce.com and Google, spanning philanthropy, business applications, social networks, and cloud computing.

- "Developers now can take advantage of the easy to use and rapidly scalable cloud computing infrastructures from Google and Salesforce.com to build and deliver powerful business applications.

- "At Google, as at Salesforce.com, we are committed to enabling developer success with cloud computing," said Tom Stocky, director of product management at Google.

- "Bringing together Google App Engine and **Force.com will foster the creation of new Web applications** and further demonstrate the power of the Web as a platform."

Developing Applications-Google

- Force.com for Google App Engine provides a set of tools and services meant to foster the creation of new kinds of web and business applications built and delivered entirely in the cloud.
- Instead of managing and maintaining their own client/server infrastructure, developers can use cloud computing infrastructure from Google and Salesforce.com to build, run, and deliver new applications on the Web.
- Web applications developed on App Engine are easy to build, easy to maintain, and easy to scale as traffic and data storage needs grow. App Engine offers a developer community and libraries to power consumer oriented interactive web applications.
- Force.com provides developers a complete environment to quickly build business applications that run on Salesforce.com's trusted global infrastructure. Now, consumer-oriented web applications built on App Engine can leverage enterprise data stored in Force.com.
- For end users of these applications, the integration of Force.com and Google App Engine means that applications can be powered by both systems.
- Using Force.com for Google App Engine, developers can build applications that span both Salesforce.com and Google's cloud computing platforms, and take advantage of the sum of their features as desired.

Developing Applications-Google

Force.com for Google App Engine provides:

- A means to leverage Python in a scalable cloud environment and interact directly with database, workflow, and logic capabilities in Force.com.
- Force.com for Google App Engine enables the creation of Python libraries that, when placed on Google App Engine, allow App Engine apps to read and write to Force.com using the Force.com API.
- App Engine developers get access to Force.com services and capabilities including mobile, analytics, security and sharing models, user authentication, Multilanguage and currency support, and more.
- Enterprise partner. “Force.com for Google App Engine brings together the leaders in cloud computing to help address those needs and allows corporate IT to easily scale up or down to meet changing business conditions.”
- The companies have been working together beginning in 2003 with joint work around the creation of Google.org, the alliance has been extended to work together on mashups, Salesforce for Google AdWords, Group Edition featuring Google AdWords, OpenSocial, Salesforce for Google Apps, Force.com for Google Data APIs, and now Force .com for Google App Engine.

Developing Applications-Google

- More than 5,000 Salesforce.com customers are using Google Apps, and more than 10,000 are using Salesforce for Google AdWords. And five of the top ten most popular applications on the Force.com AppExchange marketplace come from Google-related partners.

- Force.com for Google App Engine helps developers create new web and business applications that leverage the capabilities of App Engine, at the same time allowing access to Force.com data and APIs from directly within App Engine projects and applications.

Force.com for App Engine includes :

- Getting started guide
- Python library documentation
- Examples showing Python code accessing Force.com
- Testing harness for the provided library
- Wiki FAQ page on developer.force.com with best practices and latest tips and trick

Developing Applications-Google

Google Gears :

- Another development tool that **Google offers is Google Gears,** an open-source technology for creating offline web applications. This browser extension was made available in its early stages so that the development community could test its capabilities and limitations and help Google improve upon it.

- **Google's long-term hope is that Google Gears** can help the industry as a whole move toward a single standard for offline capabilities that all developers can use.

- Google Gears addresses a major user concern: availability of data and applications when there's no Internet connection available, or when a connection is slow or unreliable.

- "With Google Gears we're tackling a key limitation of the browser in order to make it a stronger platform for deploying all types of applications and enabling a better user experience in the cloud," said Eric Schmidt, CEO of Google.

- Google offers Google Gears as a free, fully open-source technology in order to help every web application, not just Google applications. As an example of what is possible, the Google Reader feed reader (http://reader.google.com) is available with Gears-enabled offline capabilities.

Developing Applications-Google

Sample Applications – Gears API – Google Code

http://code.google.com/apis/gears/sample.html

Google Code

Search

English | Site Directory

Gears API

Home | Docs | FAQ | Blog | Group | Terms | Contribute

Install Google Gears
Sample Applications

Developer's Guide
- Getting Started
- Tutorial
- Architecture
- Security
- Resources and Tools
- Gears on Mobile Devices

API Reference
- Summary
- Factory
- Blob
- Database
- Desktop
- Geolocation
- HttpRequest
- LocalServer
- Timer
- WorkerPool

Upcoming Release
- Developer's Guide
- API Reference

Sample Applications

This page lists sample applications that use Gears API. Sample code is found in the /samples directory of the the zipfile provided below, or by using the *view source* feature of your browser.

Download the sample code from the Resources and Tools page.

"Hello, World" Samples

Note : To see the sample applications working, you must first install Gears.

Database Demo
Shows how to read and write data locally using the Database module.

File System Demo
Demonstrates using the file picker for choosing multiple files; filtering on MIME type; and displaying the results to the screen

Geolocation Demo
Demonstrates basic usage of the Geolocation module.

ResourceStore Demo
Uses the LocalServer module to capture a URL so that it can be viewed while offline.

ManagedResourceStore Demo
Uses the LocalServer module to capture a versioned set of URLs, and updates the entire set to a new version.

Shortcut Demo
Shows how to create desktop shortcuts using the Desktop module.

WorkerPool Demo
Performs a long-running computation without blocking the UI, using the WorkerPool module.

Sample Application: Gearpad

Gearpad is simple web-based notepad which allows you to read and write notes, even offline. Gearpad transitions between online/offline states seamlessly, even when the network connection is intermittent.

Source location: /samples/gearpad

Modules used: Database, LocalServer.

To run Gearpad:
- Set up a PHP server and a MySQL database
- Create a database using the db.sql script
- Place the Gearpad files under you PHP server's document root
- Customize _dbconfig.php if necessary

Sample Application: RunningMan

A simple JavaScript application that uses the Gears APIs on Android. RunningMan is a location-aware stopwatch that measures both the time and route taken for a journey, showing the journey on a map.

Check out the RunningMan tutorial for step-by-step instructions.

Modules used: Database, Desktop shortcuts, Geolocation, and LocalServer.

Done

Developing Applications-Google

In this Google Gears application, some searched phrases are maintained on the local machine.

Developing Applications-Google

- Google intends to work closely with all members of the web community to converge upon a standard so developers have one consistent API for offline functionality.
- "The Gears API will also be available in Apollo, which enables web applications to run on the desktop, providing developers with consistent offline and local database solutions."
- "Opera and Google share the common goal of making Web applications richer and more robust," said Håkon Wium Lie, CTO, Opera Software. "Developers have long desired the functionality and flexibility Google Gears can offer browsers.
- Opera has always prioritized giving our users what they want, we're excited to work with Google to extend the reach and power of Web applications."
- Google Gears builds on the Web's existing programming model by introducing new JavaScript APIs for sophisticated data storage, application caching, and multithreading features.
- With these APIs, developers can bring offline capabilities to even their most complex web applications. Google Gears works with all major browsers on all major platforms: Windows, Mac, and Linux. A sample of Google Gears in use is
- Google Gears is available at http://gears.google.com.

Developing Applications-Microsoft

- **Microsoft's Azure Services Platform** is a tool provided for developers who want to write applications that are going to run partially or entirely in a remote datacenter.
- The **Azure Services Platform (Azure) is an Internet-scale** cloud services platform hosted in Microsoft datacenters, which provides an operating system and a set of developer services that can be used individually or together.
- **Azure can be used to build new applications** to run from the cloud or to enhance existing applications with cloud-based capabilities, and it forms the foundation of all Microsoft's cloud offerings.
- Its open architecture gives developers the choice to build web applications, applications running on connected devices, PCs, servers, or hybrid solutions offering the best of online and on-premises.
- Azure allows **developers to quickly create applications running in the cloud** by using their existing skills with the **Microsoft Visual Studio development environment and the Microsoft .NET Framework.**
- In addition to managing code languages supported by .NET, Microsoft plans more support for additional programming languages and development environments.

Developing Applications-Microsoft

Live Services

Home | Services | Resources | Community | About

Quick Applications (beta)

Quick Applications offer customizable out-of-the box solutions for specific Web scenarios. Each Quick App is built on Live Services and is offered as a source code download you can use right away.

Download Quick Apps Beta Now

e-Commerce: Retail

Retail uses Silverlight 2 and Silverlight Streaming to power an e-commerce site with product videos offering an engaging user interface.

Tour the Demo Site
Download the code

Sports: Field Manager

Field Manager allows people who play sports to join a league and keep track of different sporting fields. It allows the "player" to keep up to date as to where the field is, if it is open, and what the playing conditions are.

Tour the Demo Site
Download the code

Community: Team Builder

Team Builder allows members to see all that's happening with a specific group/team. Any member of the site can add photos, upload video and start an online conversation with other members.

Tour the Demo Site
Download the code

Events: Contoso Riders

Contoso Riders allows bicyclists to share their courses, videos and photos. Members can also set up events and chat with other members through Windows Live Messenger.

Tour the Demo Site
Download the code

Search Visualization: Tafiti

Tafiti Search Visualization uses the Windows Live Services to provide a compelling search experience for users and allow them to share that experience with others.

Tour the Demo Site
Download the code

Hospitality: Adventure Works Resorts

Connects travel and hospitality firm with customers to enrich their trip experiences.

Tour the Demo Site
Download the code

Microsoft's Azure offers a number of applications that you can use right away.

Developing Applications-Microsoft

- Infrastructure management is automated with a platform that is designed for high availability and dynamic scaling to match usage needs with the option of a **pay-as-you-go pricing model.**

- Azure provides an open, standards-based, and interoperable environment with support for multiple Internet protocols, including **HTTP, REST, SOAP, and XML.**

- Microsoft also offers cloud applications ready for consumption by customers such as Windows Live, Microsoft Dynamics, and other Microsoft Online Services for business such as Microsoft Exchange Online and SharePoint Online.

- The Azure Services Platform lets developers provide their own unique customer offerings by offering the foundational components of compute, storage, and building block services to author and compose applications in the cloud.

- Azure utilizes several other Microsoft services as part of its platform, known as the **Live Mesh platform.**

Developing Applications-Microsoft

Live Services :

- Live Services is a set of building blocks within the Azure Services Platform that is used to handle user data and application resources.

- Live Services provides developers with a way to build social applications and experiences across a range of digital devices that can connect with one of the largest audiences on the Web.

Microsoft SQL Services :

- Microsoft SQL Services enhances the capabilities of Microsoft SQL Server into the cloud as a web-based, distributed relational database.

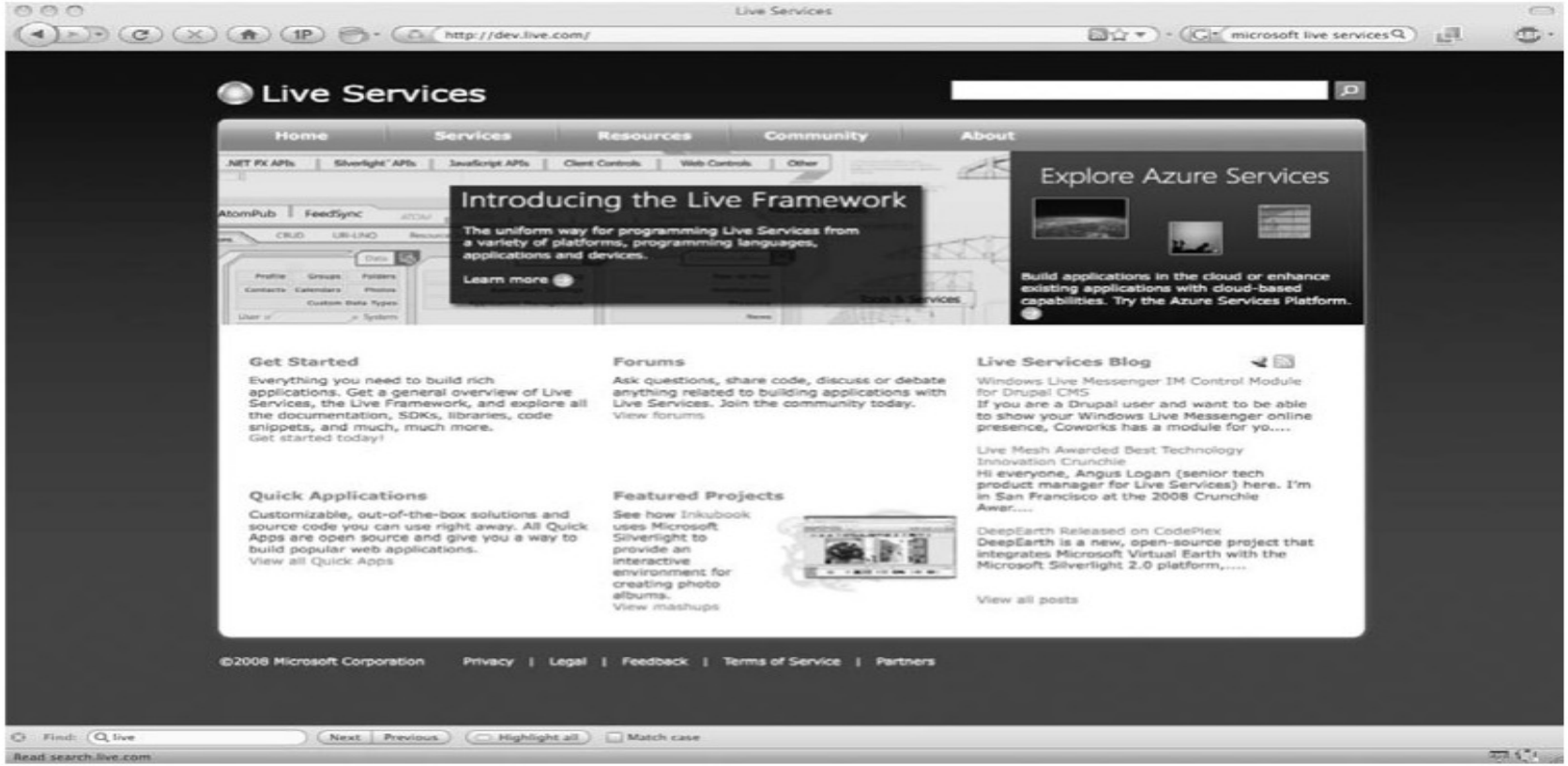
- It provides web services that enable relational queries, search, and data synchronization with mobile users, remote offices, and business partners. It can store and retrieve structured, semistructured, and unstructured data.

Microsoft .NET Services :

- Microsoft .NET Services is a tool for developing loosely coupled cloud-based applications. .NET Services includes access control to help secure applications, a service bus for communicating across applications and services, and hosted workflow execution.

- These hosted services allow the creation of applications that span from on-premises environments to the cloud

Developing Applications-Microsoft



Microsoft SharePoint Services and Dynamics CRM Services :

- **Microsoft SharePoint Services and Dynamics CRM** Services are used to allow developers to collaborate and build strong customer relationships.
- Using tools like Visual Studio, developers can build applications that utilize SharePoint and CRM capabilities.
- Design Azure is designed in several layers, with different things going on under the hood.

Layer Zero :

- Layer Zero is **Microsoft's Global Foundational Service.** GFS is akin to the **hardware abstraction layer (HAL)** in Windows. It is the most basic level of the software that interfaces directly with the servers.

Layer One :

- **Layer One is the base Azure operating system**. It used to be code-named **"Red Dog,"** and was designed by a team of operating system experts at Microsoft.

- Red Dog is the technology that networks and manages the Windows Server 2008 machines that form the Microsoft hosted cloud.

Developing Applications-Microsoft

Red Dog is made up of four pillars:

- Storage (a file system)
- The fabric controller, which is a management system for deploying and provisioning
- Virtualized computation/VM
- Development environment, which allows developers to emulate Red Dog on their desktops.
- Red Dog is designed by Microsoft such that it only has to be deployed on a single machine, and then multiple instances of it can be duplicated to the rest of the machines in the cloud.

Layer Two :

- Layer Two **provides the building blocks that run on Azure**. These services are the aforementioned Live Mesh platform. Developers build on top of these lower-level services when building cloud apps.

- SharePoint Services and CRM Services are not the same as SharePoint Online and CRM Online. They are just the platform basics that do not include user interface elements.

Layer Three :

- At Layer **Three exist the Azure-hosted applications**. Some of the applications developed by Microsoft include SharePoint Online, Exchange Online, Dynamics CRM, and Online. Third parties will create other applications.

Intuit QuickBase

- Intuit Inc.'s **QuickBase launched its new QuickBase** Business Consultant Program. The program allows members to use their expertise to create unique business applications tailored specifically to the industries they serve without technical expertise or coding.

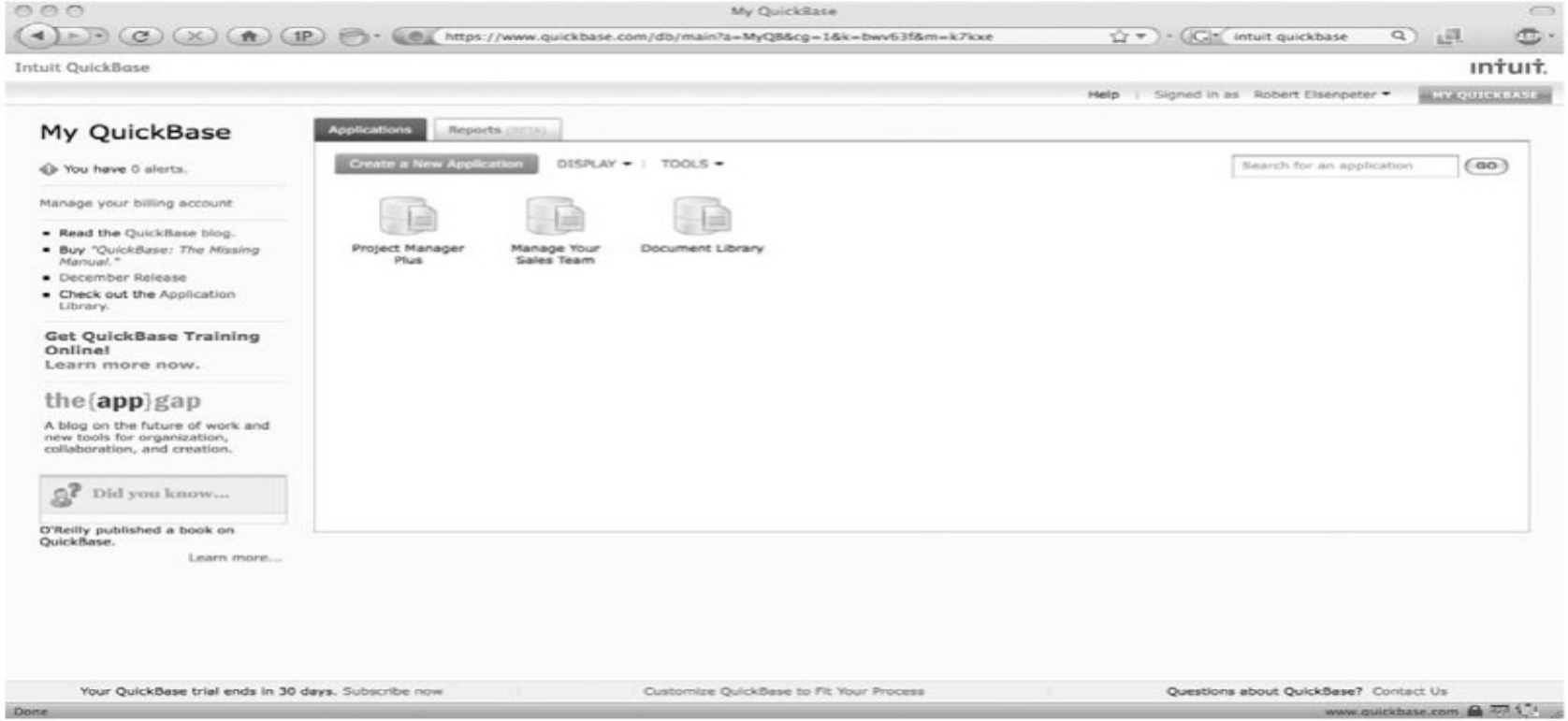
- This helps members expand their reach into industries formerly served only by IT experts. Using QuickBase, program members will be able to **easily build new on-demand business applications** from scratch or customize one of 200 available templates and resell them to their clients.

- "SaaS implementations do not require highly technical people, although they may require consultants with data integration skills.

- Instead, seasoned business process management consultants will become the focus for successful project delivery," according to Forrester Research Inc.'s Oct. 29, 2008 report, "SaaS Economics Will Change ISVs' SI And VAR Channels."

- Before joining the QuickBase program, Coates' firm was limited to recommending off-the-shelf applications that cost more than $100,000 or custom-developed solutions built on other platforms.

Intuit QuickBase



Intuit QuickBase

- A **QuickBase application customized** specifically for a customer's business process in a quarter of the time, at half the cost and twice the margin," said Coates.

- It's a software-as-a-service application, the costs are lower for my customers and they can be up and running in weeks, rather than months."

- The program provides training, partner relationship management, and lead-generation tools to help them locate potential customers and maximize the power of QuickBase.

- QuickBase Business Consultants also receive a free version of QuickBooks Online to help them better manage and grow their own businesses.

- Creating a tailored solution for business on a SaaS platform is not only possible, it's easy," said Bill Lucchini, vice president and general manager of QuickBase. "Making software fit the business needs was formerly the exclusive territory of IT and engineers.
- The QuickBase Business Consultant Program, the value shifts to the entrepreneurs and industry experts who can now solve real business problems without being a technical expert. Entrepreneurs and forward-thinking VARs will see huge benefits from this shift. **http://quickbase.intuit.com/links/partners.asp.**

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Cast Iron Cloud

- **Cast Iron Systems introduced its development platform**, the Cast Iron Cloud. Cast Iron offers the choice of a completely cloud-based integration service or an on-premise integration appliance as an organization's application ecosystem evolves.

- Any organization, regardless of size or resources, can connect SaaS solutions with other on-demand and on-premise applications, immediately boosting productivity.

- Companies must be able to efficiently integrate SaaS solutions with other enterprise systems to effectively orchestrate cross-functional business processes.

- By delivering our integration solution in the Cast Iron Cloud, or via our on-premise appliances, we can provide organizations with a seamless and secure transition between on-demand and on-premise environments."

- As SaaS usage expands from departmental silos into the extended enterprise, integration of data and applications is even more critical to productivity and success.

Cast Iron Cloud

- Cast Iron and its partners can deliver the most widely used solution for connecting SaaS and enterprise applications through the simplicity and speed of Integration as a Service (IaaS).
- The Cast Iron Cloud leverages the company's delivery of completed integration projects quickly and also eliminates the need for customers to invest in integration infrastructure or deep middleware expertise.
- "Our customers have been very successful using Cast Iron to bring Force.com and Salesforce CRM into their IT environments," said Ariel Kelman, senior director of platform product marketing at Salesforce.com.
- "With Cast Iron integration solutions available on the Force.com AppExchange, small businesses and large enterprises alike tell us that Cast Iron gives them great time to value for their integration projects."
- "Given the growing number of vertical solutions built on NS-BOS for NetSuite, as well as the increasing sophistication of our customer base, we wanted to provide our customers and partners options on how to connect to specific vertical and legacy systems.
- "With built-in connectivity to NetSuite solutions and the flexibility to deliver integration in the cloud or using an appliance, Cast Iron accelerates adoption of our ERP, ecommerce, and CRM suite."

Cast Iron Cloud

Mozilla Firefox

http://www.castiron.com/integration-solutions/cloud/studio_pop.html

File Edit View Project Orchestration Map Tools Solutions Help

Configuration-Wizard

Configuration

1. Introduction
2. Enter Credentials
3. Verify Salesforce.com connectivity
4. Get custom fields
5. Configure Polling Interval
6. **Change Account Mapping**
7. Verify Integration

Change Account Mapping

You may update the map based on any custom fields you might have added to Salesforce.com.

Design | Test

From Orchestration — Select Inputs...

Salesforce.com_Account
- Account
  - Id
  - IsDeleted
  - MasterRecordId
  - Name
  - Type
  - ParentId
  - BillingStreet
  - BillingCity
  - BillingState
  - BillingPostalCode
  - BillingCountry
  - ShippingStreet
  - ShippingCity
  - ShippingState
  - ShippingPostalCode
  - ShippingCountry
  - Phone

To Orchestration — Select Outputs...

NetSuite_Customer
- Customer
  - @internalId
  - @externalId
  - customForm
    - @internalId
    - @externalId
    - @type
    - name
  - entityId
  - isPerson
  - salutation
  - firstName
  - middleName
  - lastName
  - companyName
  - entityStatus
    - @internalId
    - @externalId

< Back | Next > | Finish | Cancel | Help

Login

Close Window

Done

Cast Iron Cloud

- Analyst firms Saugatuck Technology and Gartner explain the importance of this announcement for both SaaS users and providers. "To take advantage of SaaS at an enterprise level, users must have access to broad, deep, and sophisticated integration capabilities.

- Requirements for integration could stifle effective and efficient enterprise-wide deployment of SaaS unless SaaS providers commit to delivering more and better enterprise-ready integration solutions.

- According to Gartner, "An analysis of a cross-section of successful SaaS vendors reveals that business-to-business (B2B) flexibility rather than any one B2B standard or approach lowers barriers to SaaS integration and helps drive SaaS adoption.

- The diversity of SaaS integration solutions from SaaS vendors is good because it gives customers more options to solve their diverse integration requirements."

- Cast Iron is anticipating these diverse end-to-end application integration needs, including data cleansing and migration, and application integration by offering intelligent connections, data transformations, process workflow, monitoring and management, plus guaranteed delivery of all data.

Cast Iron Cloud

- Cast Iron is transforming the integration experience using the Cast Iron Cloud. The company is introducing a cloud-based library of preconfigured Template Integration Processes (TIPs) for the most common SaaS business processes.
- Cast Iron has created these templates based on its experience with thousands of customer integrations.
- For example, if customers need to integrate two SaaS applications, they simply search Cast Iron's cloud based library of TIPs, choose the TIP that matches their scenario, and deploy it to the Cast Iron Cloud.
- In minutes, their SaaS integration project goes live rather than taking weeks or even months to develop using custom code.In addition, SaaS integrations can be monitored from anywhere at any time using the Cast Iron Cloud.
- For companies that want to customize TIPs based on their specific requirements, Cast Iron is providing a self-guided wizard similar to the simple wizard-based experience in popular products like Intuit TurboTax.
- Users answer a few questions based on the specific situation, and the integration process is automatically customized to expedite SaaS integration and adoption.

Bungee Connect

- Bungee Labs offers its Bungee Connect web application development and hosting platform. Developers use Bungee Connect to build desktop-like web applications that leverage multiple web services and databases, and then deploy them on Bungee's multitenant grid infrastructure.

- Bungee Connect provides development, testing, deployment, and hosting in a single, on-demand platform. The company claims that using their solution, significant time and cost are eliminated across the entire application life cycle reducing time-to-market by as much as 80 percent.

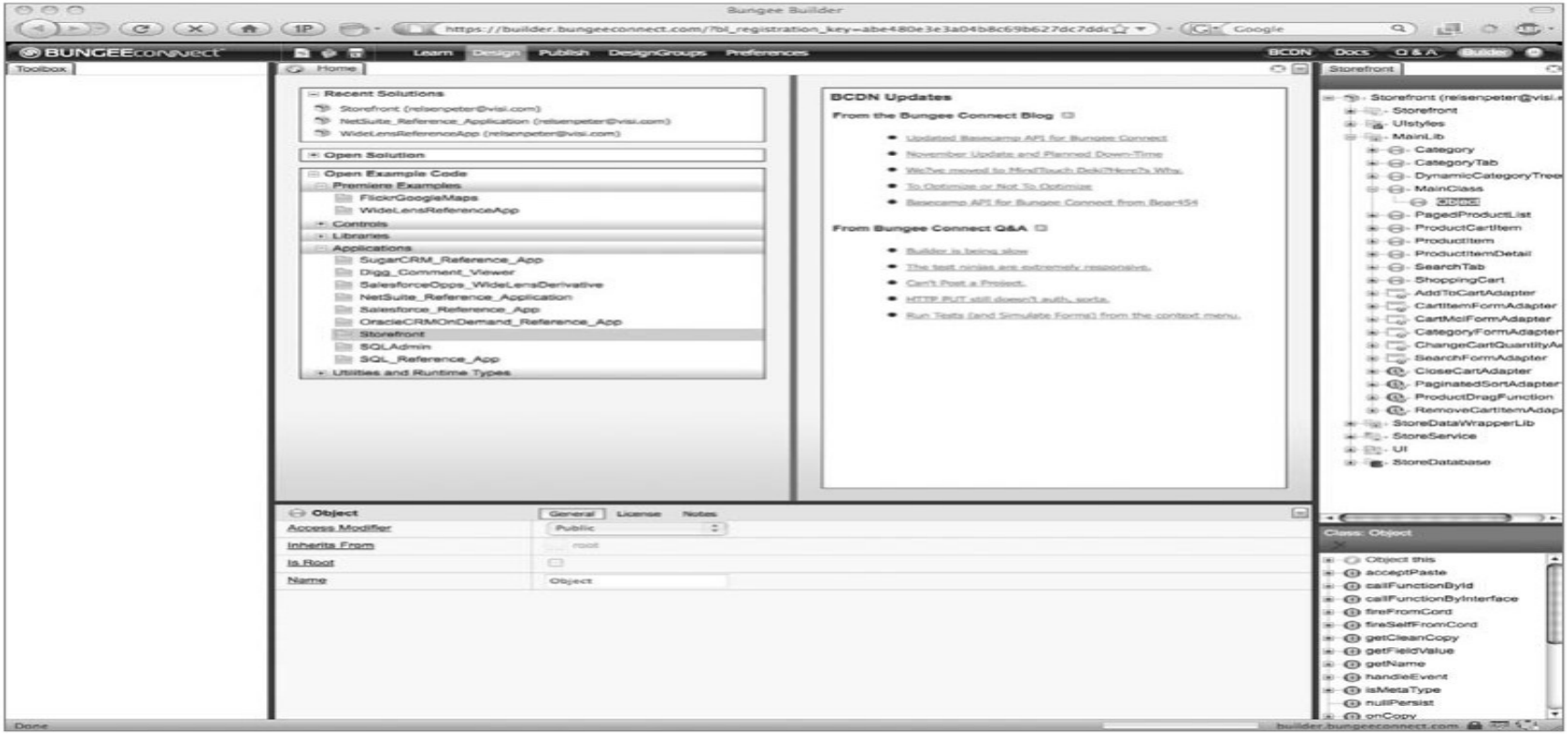
Bungee Connect includes the following features:

- A single, on-demand environment for developing, testing, deploying, and hosting
- Interaction delivered entirely via browser with no download or plug-in for developers or end users
- Delivery of highly interactive user experience without compromising accessibility and security
- Automated integration of web services (SOAP/REST) and databases (MySQL/PostgreSQL)
- Built-in team collaboration and testing
- Built-in scalability, reliability, and security
- Deep instrumentation of end-user application utilization for analytics
- Utility pricing model based on end-user application use

Bungee Connect

- Developers and IT managers can leverage a completely online platform to build and deliver powerful, AJAX-enabled, multiple data-source web applications.
- Applications may be embedded within other web applications and pages, Software as a Service (SaaS) solutions, or delivered as stand-alone web destinations.
- "Cloud computing, internal-external mashups and online business services are hastening the need for new business models that can support entire application lifecycles," said Dana Gardner, principal analyst of Interarbor Solutions.
- "It's not enough to develop as a service, or to deploy as a service the fuller cost-benefit payback comes from the application lifecycle as a service.
- All the better that costs to support the applications from inception to sunset are commensurate with use and demand.
- Enterprise IT innovators and efficiency-minded independent developers alike should view the Bungee Connect model as the platform and mashup approach of the future."

Bungee Connect



Bungee Connect

- Development, team collaboration, and test deployment hosting on Bungee Connect are free of charge. Developers pay only when their applications are actually used by end users.

- Once a Bungee-powered application is deployed on the Bungee Grid infrastructure, end-user utilization of the application is deeply instrumented to provide developers with detailed insight into application performance and usage patterns, and to compute the application's utility bill.

- Depending on the usage profile of an application, businesses can expect to pay between US$2 to US$5 per user per month for a heavily used business productivity application, or fractions of a cent per e-commerce transaction.

- All Bungee-powered applications are hosted for free during the Bungee Connect Public Beta. For more information about Bungee Connect, visit http://www.bungeeconnect.com.

Development-Google App Engine

- There are many different places you can turn to develop your own applications.
- The development steps with three different, popular platforms: the Google App Engine, Salesforce.com, and Microsoft Azure.

Google App Engine :

- Google uses Python
- To get started, you must first download the Google App Engine development kit, located at http://code.google.com/appengine/downloads.html.
- The SDK is available for Windows, Mac OS X, and Linux environments that also have Python 2.5.
- The SDK includes a web server application that simulates the App Engine environment.
- It also includes a local version of datastore, Google Accounts, and the ability to get URLs and send email from your computer using the App Engine APIs.

Development-Google App Engine

- Need Python 2.5 installed on your computer. Mac OS X Leopard users already have Python 2.5 installed. You can download Python from the Python web site at http://www.python.org/.
- Download and install the App Engine SDK. For this demonstration, you will use two commands from the SDK:
- dev_appserver.py The development web server
- appcfg.py Used to upload your app to App Engine
- If you are using the Zip archive version of the SDK, those commands are located in the google_appengine director.

The App :

- App Engine applications communicate with the web server using the CGI standard. When the server receives a request for your application, it runs the app with the request data in environment variables and on the input stream.
- When it responds, the app writes the response to the output stream and includes HTTP content.

Development-Google App Engine

- Here's Johnny! The app we are going to make displays a little greeting. To get started, make a directory named heresjohnny. All the files for this app will be stored in this directory.

- Inside that directory, create a file named heresjohnny.py to include this code:

- print 'Content-Type: text/plain'
- print ''
- print 'Here's Johnny!'
- What's going on here is that the Python script responds to the request with an HTTP header that describes content (HTTP), a blank line, and the message, "Here's Johnny!

Development-Google App Engine

- Configuration File Next, you need to have a configuration file called app.yaml.

- Create a file in the directory called app.yaml and write it to read as follows:

- application: heresjohnny
- version: 1
- runtime: python
- api_version: 1
- handlers:
- - url: /.*
- script: heresjohnny.py

app.yaml

```
application: Here's Johnny
version: 1
runtime: python
api_version: 1

handlers:
- url: .*
  script: main.py
```

Development-Google App Engine

Here's what the code is doing, line by line:

- The application identifier is heresjohnny. When the application is registered with App Engine, you select a unique identifier, so this will change.It can be named whatever you want at this stage.
- This is the first version of this code, this is number 1. Use this field to keep track of different versions of your app's code.
- This code runs in the Python runtime environment, version 1.
- Eery request to a URL whose path matches the regular expression /.* will be handled by the heresjohnny.py script.
- Script Testing Because the handler script and configuration file are mapping every URL to the handler, the application is done. That's it. Now you can test the app with the web server included with the App Engine SDK.
- Start the web server with the following command (including the path to the heresjohnny directory):
- google_appengine/dev_appserver.py heresjohnny/

Development-Google App Engine

- Now that the web server is running, it is listening for requests on port 8080. Check to make sure it is using that port by entering this URL in a web browser: http://localhost:8080/
- You can leave the web server running, even if you make changes to the app. The web server knows to watch for changes in your source files and reload them if need be. Test it out. Make a change to your heresjohnny.py file and then reload the page by entering http://localhost:8080.

Uploading the App :

- After creating and managing your app and registering it on Google, the next step is to upload the application using a command-line tool included in the SDK called appcfg.py.
- Registration After you've created your app using the Administration Console (found at http://appengine.google.com/), the next step is to register the application ID for your application.

Development-Google App Engine

Go, change the world®

Create an Application

http://appengine.google.com/start/createapp

Google App Engine

robertelsenpeter@gmail.com | My Account | Help | Sign out

Create an Application

Application Identifier:

Hello Johnny! .appspot.com Check Availability Sorry, "Hello Johnny!" is not available. Application identifiers may contain only numbers, lowercase letters or hyphens and must be between 6 and 30 characters.

You can map this application to your own domain later. Learn more

Application Title:

Hello Johnny!

Displayed when users access your application.

Authentication Options (Advanced): Learn more

Google App Engine provides an API for authenticating your users. If you choose not to use this, anyone in the world will be able to access your application. However, if you choose to use this, you'll need to specify now who can sign in to your application.

Open to all Google Accounts users (default)

If your application uses authentication, anyone with a valid Google Account may sign in. (This includes all Gmail Accounts, but does "not" include accounts on any Google Apps domains.)

Edit

Terms of Service:

1. Your Agreement with Google

1.1. Your use of the Google App Engine service (the "Service") is governed by this agreement (the "Terms"). "Google" means Google Inc., located at 1600 Amphitheatre Parkway, Mountain View, CA 94043, United States, and its subsidiaries or affiliates involved in providing the Service.

☑ I accept these terms.

Save Cancel

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Done

Use the Administration Console to register your apps on Google App Engine.

Development-Google App Engine

- The app will still be called Here's Johnny. There can be a million apps named Here's Johnny, but they all need different registration codes.
- Once the registration is successful, you see a screen like the one in Figure that lets you start managing your app on the Google Cloud.
- Once the registration is completed, you access the application by going to http://application-id.appspot.com.
- In our case, it would be http://pos7412.appspot.com.
- Next, you have to go back into your app.yaml file and change the value of the application: setting. In our case, it is being changed from heresjohnny to pos7412.
- This tells the system what the app is registered as.
- Send It In To upload your finished application to Google App Engine, run the following :
- command:appcfg.py update heresjohnny/

Development-Google App Engine

- Enter your Google username and password at the prompts.
- Now you can see your application on App Engine and all you need to do is open up a web browser and enter http ://application-id.appspot.com.
- You can do much more with the Google App Engine SDK. You can find out more information on development from whatever vendor you get an SDK for.

Once you've registered the app, Google lets you know that it's got a home.

Development- Salesforce.com

- Salesforce has two ways to create an application—using point-and-click methods or by using the command line. The GUI method of application developing an application :

Create an Account :

- The first step is to create your Salesforce account. This is a fairly straightforward affair and it's something you've probably already done on dozens of web sites.

- You need to go to http://developer.force.com/join. Simply follow the steps (you'll need to answer name, password, username, and so forth) and you'll be set up.

- Once you log in to your Developer Edition organization, then log in to https://login.salesforce.com.

Create an Object :

- The next step is to create an object that will hold the data—our lunch expenses. Salesforce applications already contain a lot of prebuilt object called standard object for the data your applications need.

- However, you can build custom applications. Those are called custom objects.

Development- Salesforce.com

1. Requirement Analysis

- Understand business needs and processes
- Identify users, data, security, and automation requirements
- Decide whether standard Salesforce features are sufficient or customization is needed

2. Choose Salesforce Cloud & Edition

- Select the appropriate cloud: **Sales Cloud, Service Cloud, Experience Cloud, Marketing Cloud**, etc.
- Choose the edition (Essentials, Professional, Enterprise, Unlimited) based on scalability and features

3. Org Setup

- Create a **Salesforce Org** (Developer / Sandbox / Production)
- Configure company details, time zone, language, and currencies

4. Data Modeling

- Define **Objects** (Standard or Custom)
- Create **Fields**, relationships (Lookup, Master–Detail)
- Design page layouts and record types

5. Security & Access Control

- Define **Profiles**, **Roles**, and **Permission Sets**
- Set **Object-level, Field-level, and Record-level** security
- Configure authentication and sharing rules

6. Customization & Configuration

- Configure **Validation Rules**
- Create **Workflows**, **Process Builder**, or **Flows**
- Customize UI using Lightning App Builder

7. Apex Development (if required)

- Write **Apex classes and triggers** for complex business logic
- Implement batch jobs, schedulers, and integrations

8. UI Development

- Build user interfaces using **Lightning Web Components (LWC)** or **Aura Components**
- Customize dashboards and reports

9. Integration

- Integrate Salesforce with external systems using
- REST / SOAP APIs
- Middleware (MuleSoft, Boomi, etc.)
- Implement real-time or batch integrations

10. Testing

- Perform unit testing and system testing
- Ensure **Apex test coverage ≥ 75%**
- Validate functionality, security, and performance

11. Deployment

- Use **Change Sets**, **Salesforce CLI**, or **DevOps tools**
- Deploy from Sandbox to Production

12. User Training & Documentation

- Train end users and administrators
- Prepare user guides and technical documentation

13. Maintenance & Support

- Monitor system performance
- Fix bugs, enhance features, and apply Salesforce updates

Development- Salesforce.com

1. Requirement Analysis
↓
2. System Design
↓
3. Development
↓
4. Testing
↓
5. Deployment
↓
6. Maintenance

CI/CD in Salesforce – Key Pointers :

- **CI (Continuous Integration)**
  Automatically integrates code changes from multiple developers into a shared repository.

- **Version Control**
  Uses Git (GitHub / GitLab / Bitbucket) to manage Salesforce metadata and source code.

- **Build Automation**
  Salesforce CLI retrieves, converts, and validates metadata automatically.

- **Automated Testing**
  Runs Apex unit tests to ensure ≥ **75% code coverage** and code quality.

- **CD (Continuous Deployment)**
  Automatically deploys validated changes to Sandbox / Production.

- **Deployment Tools**
  Uses Change Sets, Salesforce CLI, DevOps Center, or tools like Jenkins, GitHub Action

Development- Salesforce.com

- To create the object, log in to your Salesforce account at https:/login.salesforce.com and enter your username and password. create the custom object by following these steps:

- Select Setup in the upper-right corner of the page.

- Click Create Objects in the sidebar menu to show the Custom Objects page.

- Fill in the custom object definition. This is shown in Figure 11-5:
- Under Label, enter Lunch.
- Under Plural Label, enter Lunch.
- Under the Object Name, enter Lunch.
- Under the Description, enter Anobjectthatholdslunchexpenseinformation.
- Select the Allow Activities check box.
- Select the Allow Reports check box.
- Leave all the other values at their default levels.
- Click Save to finish the custom object creation.

Development- Salesforce.com

Create – Salesforce – Developer Edition

https://na6.salesforce.com/ui/setup/Setup?setupid=DevTools&page=Setup&retURL

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- Manage your custom objects
- Share apps and components with other users via Force.com AppExchange
- Manage your custom report types
- Manage your custom tabs
- Customize your Workflow Rules
- Customize your Approval Processes

Getting Started
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New Custom Object — Help for this Page

Custom Object Definition Edit — Save | Save & New | Cancel

Custom Object Information — * = Required Information

The singular and plural labels are used in tabs, page layouts, and reports.

Label: Lunch — Example: Account

Plural Label: Lunch — Example: Accounts

The Object Name is used when referencing the object via the API.

Object Name: Lunch — Example: Account

Description: Anobjectthatholdslunchexpenseinformation

Context-Sensitive Help Setting:
- Open the standard Salesforce Help & Training window
- Open a window using a custom s-control

Custom S-Control: --None--

Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

Record Name: Lunch Name — Example: Account Name

Data Type: Text

Optional Features
- [x] Allow Reports
- [x] Allow Activities
- [ ] Track Field History

Deployment Status — What is this?
- In Development
- Deployed

Object Creation Options (Available only when custom object is first created)
- [ ] Add Notes and Attachments related list to default page layout
- [ ] Launch New Custom Tab Wizard after saving this custom object

Save | Save & New | Cancel

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FIGURE 11-5 Start creating your app by defining a custom object.

Fields :

- After the custom object is made to hold your lunch expenses, the next step is to add fields that capture the date of your business lunch, how much you spent, and the person you entertained.
- **Date First, create the Date field (shown in Figure 11-6):**
- Scroll down to the Custom Fields & Relationships related list.
- Select New to start the New Custom Field Wizard.
- Under Data Type, select Date and click Next
- **Enter the following data (shown in Figure 11-7):**
- Under Field Label, enter Date.
- Under Field Name, enter Date.
- Under the Description, enter Dateoflunch.
- Check the Required check box.
- For the Default Value, enter Today().
- Click Next, accept the defaults, and click Next again.
- Click Save & New to create the Date field and to restart the wizard for the next field

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Lunch
New Custom Field

Help for this Page

Step 1. Choose the field type — Step 1

Next | Cancel

Specify the type of information that the custom field will contain.

Data Type	
None Selected	Select one of the data types below.
Auto Number	A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.
Formula	A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.
Roll-Up Summary	A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.
Lookup Relationship	Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.
Master-Detail Relationship	Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where: - The relationship field is required on all detail records. - Once the value of the relationship field has been saved, it cannot be changed. - The ownership and sharing of a detail record are determined by the master record. - When a user deletes the master record, all detail records are deleted. - You can create rollup summary fields on the master record to summarize the detail records. The relationship field allows users to click on a lookup icon to select a value from a popup list. The master object is the source of the values in the list.
Checkbox	Allows users to select a True (checked) or False (unchecked) value.
Currency	Allows users to enter a dollar or other currency amount and automatically formats the field as a currency amount. This can be useful if you export data to Excel or another spreadsheet.
Date	Allows users to enter a date or pick a date from a popup calendar.
Date/Time	Allows users to enter a date and time, or pick a date from a popup calendar. When users click a date in the popup, that date and the current time are entered into the Date/Time field.
Email	Allows users to enter an email address. The entered address is validated to ensure that it is in the proper format. Users can then click on the field to automatically launch their email program and send an email to that address.
Number	Allows users to enter any number. Leading zeros are removed.
Percent	Allows users to enter a percentage number, for example, "10" and automatically adds the percent sign to the number.
Phone	Allows users to enter any phone number. Automatically formats it as a phone number.
Picklist	Allows users to select a value from a list you define.
Picklist (Multi-Select)	Allows users to select multiple values from a list you define.
Text	Allows users to enter any combination of letters and numbers.
Text Area	Allows users to enter up to 255 characters on separate lines.
Text Area (Long)	Allows users to enter up to 32000 characters on separate lines.
URL	Allows users to enter any valid website address. When users click on the field, the URL will open in a separate browser window.

Next | Cancel

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FIGURE 11-6 A key piece of data to be tracked by this app is the date.

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Lunch
New Custom Field

Help for this Page

Step 2. Enter the details — Step 2 of 4

Previous | Next | Cancel

Field Label	Date
Field Name	Date
Description	Datecflunch
Help Text	
Required	☑ Always require a value in this field in order to save a record
Default Value	Show Formula Editor Today()

Use formula syntax: e.g., Text in double quotes: "hello", Number: 25, Percent as decimal: 0.10, Date expression: Today() + 7

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FIGURE 11-7 When filled, the values in the Date field should look like this.

- **Cost** The next field to generate is the Cost field. This is where the cost of your business lunch is stored. The wizard was automatically restarted in the previous section.

- For the Data Type, select Number (shown in Figure 11-8).

- Fill in the fields as follows (shown in Figure 11-9):

1. Under Field Label, enter Cost.
2. For the Length, enter 4.
3. Under Decimal Places, enter 2.
4. For the Field Name, enter Cost.
5. For the Description, enter Costoflunch.
6. Check the Required check box.
7. Click Next, accept the defaults, and click Save & New to create the next field.

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Lunch
New Custom Field

Help for this Page

Step 1. Choose the field type — Step 1

Next | Cancel

Specify the type of information that the custom field will contain.

Data Type	
None Selected	Select one of the data types below.
Auto Number	A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.
Formula	A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.
Roll-Up Summary	A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.
Lookup Relationship	Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.
Master-Detail Relationship	Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where: - The relationship field is required on all detail records. - Once the value of the relationship field has been saved, it cannot be changed. - The ownership and sharing of a detail record are determined by the master record. - When a user deletes the master record, all detail records are deleted. - You can create rollup summary fields on the master record to summarize the detail records. The relationship field allows users to click on a lookup icon to select a value from a popup list. The master object is the source of the values in the list.
Checkbox	Allows users to select a True (checked) or False (unchecked) value.
Currency	Allows users to enter a dollar or other currency amount and automatically formats the field as a currency amount. This can be useful if you export data to Excel or another spreadsheet.
Date	Allows users to enter a date or pick a date from a popup calendar.
Date/Time	Allows users to enter a date and time, or pick a date from a popup calendar. When users click a date in the popup, that date and the current time are entered into the Date/Time field.
Email	Allows users to enter an email address. The entered address is validated to ensure that it is in the proper format. Users can then click on the field to automatically launch their email program and send an email to that address.
Number	Allows users to enter any number. Leading zeros are removed.
Percent	Allows users to enter a percentage number, for example, "10" and automatically adds the percent sign to the number.
Phone	Allows users to enter any phone number. Automatically formats it as a phone number.
Picklist	Allows users to select a value from a list you define.
Picklist (Multi-Select)	Allows users to select multiple values from a list you define.
Text	Allows users to enter any combination of letters and numbers.
Text Area	Allows users to enter up to 255 characters on separate lines.
Text Area (Long)	Allows users to enter up to 32000 characters on separate lines.
URL	Allows users to enter any valid website address. When users click on the field, the URL will open in a separate browser window.

Next | Cancel

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FIGURE 11-8 The next field to define is the Cost field.

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Lunch
New Custom Field

Help for this Page

Step 2. Enter the details — Step 2 of 4

Previous | Next | Cancel

Field Label: Cost

Please enter the length of the number and the number of decimal places. For example, a number with a length of 8 and 2 decimal places can accept values up to "12345678.90".

Length: 4 — Number of digits to the left of the decimal point

Decimal Places: 2 — Number of digits to the right of the decimal point

Field Name: Cost

Description: Costoflunch

Help Text:

Required: ☑ Always require a value in this field in order to save a record

Unique: ☐ Do not allow duplicate values

External ID: ☐ Set this field as the unique record identifier from an external system

Default Value: Show Formula Editor

Use formula syntax, e.g., Text in double quotes: "hello", Number: 25, Percent as decimal: 0.15, Date expression: Today() + 7

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FIGURE 11-9 When filled, information in your Cost field will look like this.

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New Custom Field

Help for this Page

Step 1. Choose the field type — Step 1

Next | Cancel

Specify the type of information that the custom field will contain.

Data Type	
None Selected	Select one of the data types below.
Auto Number	A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.
Formula	A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.
Roll-Up Summary	A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.
Lookup Relationship	Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.
Master-Detail Relationship	Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where: - The relationship field is required on all detail records. - Once the value of the relationship field has been saved, it cannot be changed. - The ownership and sharing of a detail record are determined by the master record. - When a user deletes the master record, all detail records are deleted. - You can create rollup summary fields on the master record to summarize the detail records. The relationship field allows users to click on a lookup icon to select a value from a popup list. The master object is the source of the values in the list.
Checkbox	Allows users to select a True (checked) or False (unchecked) value.
Currency	Allows users to enter a dollar or other currency amount and automatically formats the field as a currency amount. This can be useful if you export data to Excel or another spreadsheet.
Date	Allows users to enter a date or pick a date from a popup calendar.
Date/Time	Allows users to enter a date and time, or pick a date from a popup calendar. When users click a date in the popup, that date and the current time are entered into the Date/Time field.
Email	Allows users to enter an email address. The entered address is validated to ensure that it is in the proper format. Users can then click on the field to automatically launch their email program and send an email to that address.
Number	Allows users to enter any number. Leading zeros are removed.
Percent	Allows users to enter a percentage number, for example, "10" and automatically adds the percent sign to the number.
Phone	Allows users to enter any phone number. Automatically formats it as a phone number.
Picklist	Allows users to select a value from a list you define.
Picklist (Multi-Select)	Allows users to select multiple values from a list you define.
Text	Allows users to enter any combination of letters and numbers.
Text Area	Allows users to enter up to 255 characters on separate lines.
Text Area (Long)	Allows users to enter up to 32000 characters on separate lines.
URL	Allows users to enter any valid website address. When users click on the field, the URL will open in a separate browser window.

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FIGURE 11-10 Selecting the Lookup Relationship data type allows you to link data for two objects.

- Contact The next field is the Contact field. This is the person with whom you had your business lunch. Again, if you clicked Save & New in the previous section, the field wizard restarts.
- For the Data Type, select Lookup Relationship (shown in Figure 11-10) and click Next.
- The Lookup Relationship data type allows you to link two data objects. In this example, we'll connect the Cost object and the Contact object.
- In the Related To drop-down list, select Contact and click Next.

Fill in the field details as follows (shown in Figure 11-11):

1. Under Field Label, enter Contact.
2. Under Field Name, enter Contact.
3. Under Description, enter PersonIhadlunchwith.
4. Leave the remaining defaults as they are, and then click Next.
5. Accept the defaults again and click Next.
6. Click Next two more times.
7. Click Save to finish up the field creation process.

Create a Tab :

- To show your application on the Salesforce web site, you need to create a tab to be added to the existing tabs. When users click this tab, they'll be able to track their lunch expenses.

1. On the startup page, click Create Tabs (this was the same screen where you created the object).

2. Click New in the Custom Objects tab list to launch the New Custom Tab wizard.

3. From the Object drop-down list, select Lunch.

4. For the Tab Style, click the lookup icon and select the apple, shown in Figure 11-12 (it was our only food option).

5. Accept the remaining defaults and click Next.

6. Click Next and then Save to finish creating the tab.

7. Once the tab has been created, it is added to your set of tabs

Development- Salesforce.com

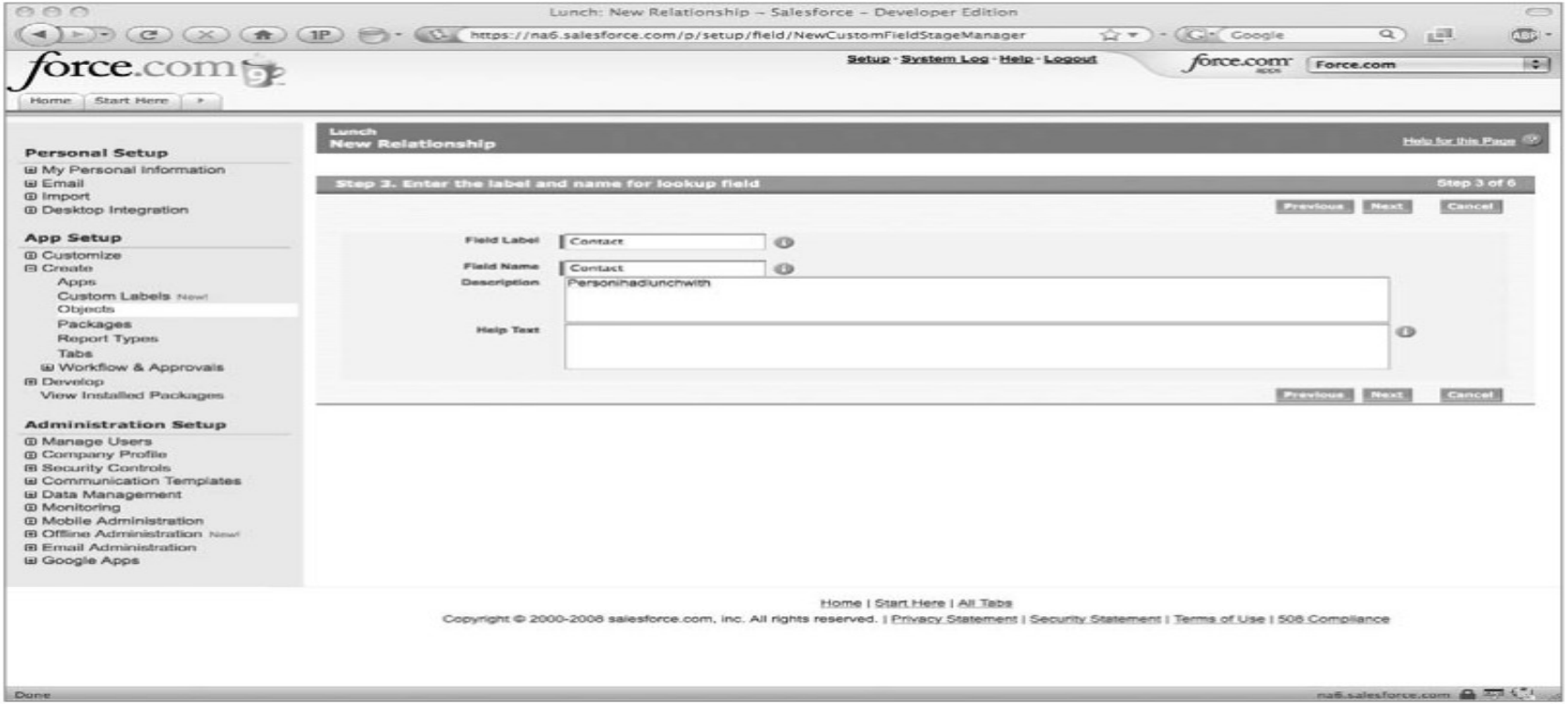

FIGURE 11-11 The Contacts field should be configured to look like this.

Make the App

- At this point, we have created three fields and a tab. These are all pulled together to create our app. You make the app by following these steps:

1. Go to Setup Create Apps.
2. Click New to launch the New Custom App Wizard.

Enter the following details (shown in Figure 11-13):

1. Under App Label, enter Lunch Tracker.
2. Under the App Name, enter Lunch_Tracker
3. Under the Description, enter Thisapplicationtracksyourlunchexpenses.
4. Click Next.
5. Accept the defaults and the default logo for the app, and then click Next.
6. In the Available Tabs box, locate the Lunch tab and click Add to add it to the selected tabs.
7. Leave the Default Landing tab set to the Home tab and click Next.
8. Select the Visible check box to make the app available to all users.
9. Click Save to create the Lunch Tracker app.
10. Repeat the steps 1–9, but this time select the Contacts tab. This is shown in Figure 11-14.
11. After the app is created, it is shown on the Force.com app menu in the upper right

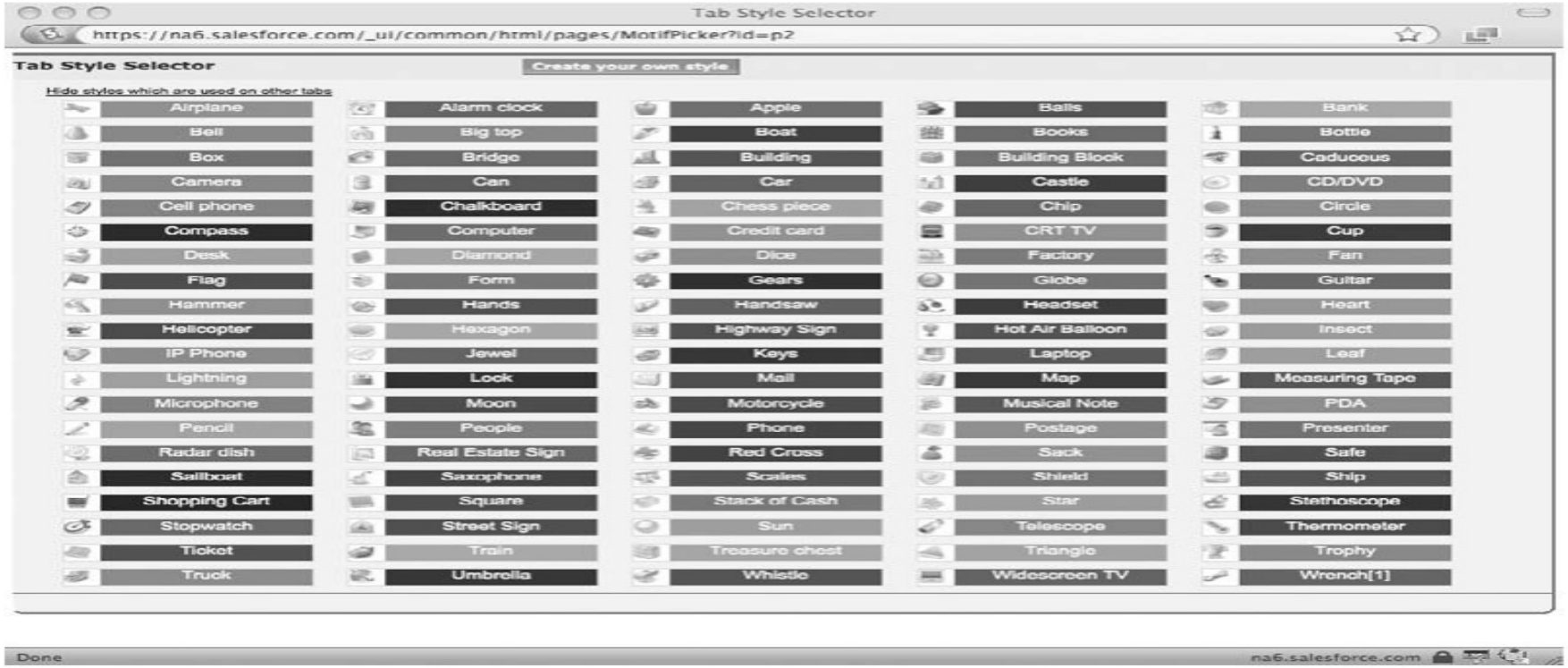

FIGURE 11-12 Start creating your app by defining a custom object.

Development- Salesforce.com

New Custom App – Salesforce – Developer Edition

https://na6.salesforce.com/ui/setup/tab/TabSetWizard?retURL=%2F02u%3FretURL

force.com

Setup · System Log · Help · Logout

Home | Start Here | Lunch

Personal Setup
- My Personal Information
- Email
- Import
- Desktop Integration

App Setup
- Customize
- Create
  - Apps
  - Custom Labels New!
  - Objects
  - Packages
  - Report Types
  - Tabs
- Workflow & Approvals
- Develop
- View Installed Packages

Administration Setup
- Manage Users
- Company Profile
- Security Controls
- Communication Templates
- Data Management
- Monitoring
- Mobile Administration
- Offline Administration New!
- Email Administration
- Google Apps

New Custom App — Help for this Page

Step 1. Enter the Details — Step 1 of 4

Fill in the fields below to define the custom app.

Custom App Information — * = Required Information

App Label: Lunch Tracker — Example: HRforce, Financeforce, Bugforce

App Name: Lunch_Tracker

Description: hisapplicationtracksyourlunchexpenses

Next | Cancel

Home | Start Here | Lunch | All Tabs

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Done — na6.salesforce.com

FIGURE 11-13 Configure your app to have settings like this one.

Test It Out :

Now that your app has been created, let's take it for a spin.

1. Select the Lunch Tracker application from the Force.com app menu at the upper right of the screen.

2. Click on the Lunch Tracker tab and click New to create a new lunch expense entry.

3. Enter some test data (our example is shown in Figure 11-15).

4. For the Contact field, click the lookup icon, click New to create a new contact named BruceDickinson, and then click Save.

5. Click Save to save your record and to return to the detail page for the new record. And there you have it.

- You were able to make your very own app on Salesforce quickly and easily, and you didn't need to use any coding—it was all point and click.

Development- Salesforce.com

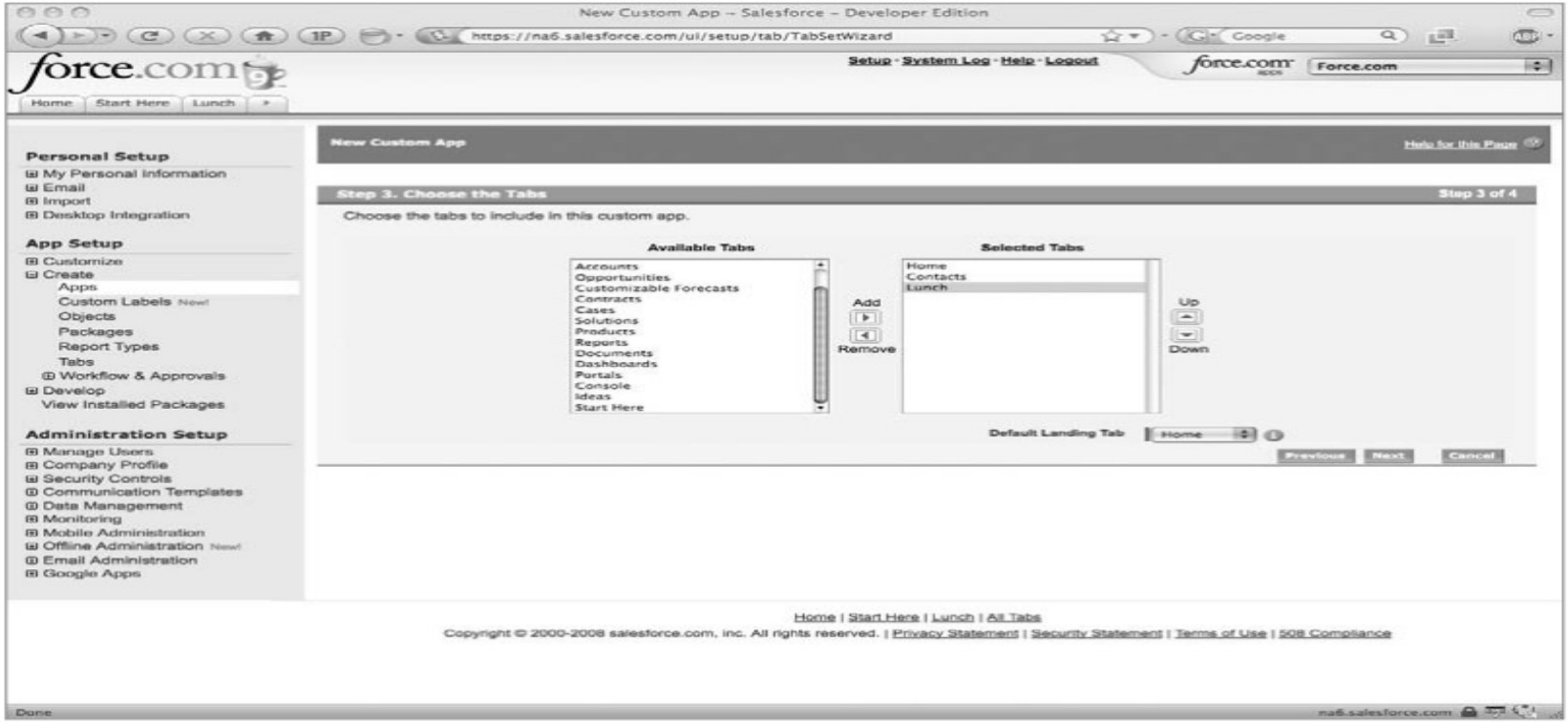

FIGURE 11-14 Choose Contacts and Lunch to be included in tabs.

Microsoft Windows Azure

- Microsoft's Azure platform is a little different to develop apps for. It is similar to Google App Engine in that you have to download an SDK, but its features are a little more rich.
- You develop applications within Visual Studio, which you then publish to the Windows cloud.
- The Azure Services Platform is designed to help you quickly and easily create, deploy, manage, and distribute web services and applications on the Internet.
- Windows Azure is an operating system for the cloud that serves as the development, run-time, and control environment for the Azure Services Platform.
- Windows Azure provides developers with on-demand compute and storage to host, scale, and manage web applications on the Internet through Microsoft datacenters.

SDK :

- To get started developing applications for Azure, you must first download the SDK.
- The January 2009 release is located at http://www.microsoft.com/downloads/details
- .aspx?FamilyID=80e3eabf-0507-4560-aeb6-d31e9a70a0a6&displaylang=en.

Application Management

Microsoft Azure Development Steps :

- **Requirement Analysis**
  Identify application needs, users, scalability, and security requirements.
- **Azure Service Selection**
  Choose suitable services (Azure App Service, VM, Functions, SQL, Storage, etc.).
- **Environment Setup**
  Create Azure account, Resource Groups, Virtual Networks, and subscriptions.
- **Application Design**
  Design architecture, data flow, APIs, and security model.
- **Development**
  Develop application using supported languages (.NET, Java, Python, Node.js).
- **Configuration & Integration**
  Configure services, connect databases, APIs, and third-party services.
- **Testing**
  Perform unit testing, load testing, and security testing.
- **Deployment**
  Deploy using Azure DevOps, CI/CD pipelines, or GitHub Actions.
- **Monitoring & Logging**
  Use Azure Monitor, Application Insights, and Log Analytics.
- **Maintenance & Scaling**
  Fix issues, update features, and scale using autoscaling and load balancers.

Application Management

Azure CI/CD :

- **Source Control** – Code stored in GitHub or Azure Repos
- **Continuous Integration (CI)** – Automatic build and unit testing
- **Build Pipeline** – Compiles code and creates artifacts
- **Automated Testing** – Unit, integration, and security tests
- **Continuous Deployment (CD)** – Deploys to Azure App Service / VM / AKS
- **Monitoring** – Azure Monitor and Application Insights

Tools Used in Azure CI/CD

- Azure DevOps
- GitHub Actions
- Azure Pipelines

Application Management

Comparison: Azure vs Salesforce vs GCP

Feature	Microsoft Azure	Salesforce	Google Cloud Platform (GCP)
Platform Type	IaaS + PaaS + SaaS	SaaS + PaaS	IaaS + PaaS
Primary Use	Cloud app hosting	CRM & enterprise apps	Cloud-native apps
Languages	.NET, Java, Python, Node	Apex, LWC	Java, Python, Go, Node
CI/CD Tools	Azure DevOps, GitHub Actions	DevOps Center, CLI, Jenkins	Cloud Build, GitHub Actions
Deployment	VM, App Service, AKS	Metadata deployment	Compute Engine, GKE
Customization	High	Medium–High	High
Best For	Enterprise cloud apps	CRM-based apps	Scalable cloud apps

Microsoft Windows Azure

Download details: Windows Azure SDK - Microsoft Internet Explorer

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Address http://www.microsoft.com/downloads/details.aspx?FamilyID=80e3eabf-0507-4560-aeb6-d31e9a70a0a6&displaylang=en

Click Here to Install Silverlight

United States Change | All Microsoft Sites | Sign out

Microsoft

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- Development Resources

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Windows Azure Software Development Kit (January 2009 CTP)

Windows

Brief Description

The Windows® Azure™ SDK provides developers with the APIs, tools, documentation, and samples needed to develop Internet-scale applications that run on Windows Azure.

On This Page

- Quick Details
- System Requirements
- Related Resources
- Overview
- Instructions
- What Others Are Downloading

Download files below

Quick Details

Version:	1.0.1.0
Date Published:	1/14/2009
Language:	English
Download Size:	62 KB - 7.3 MB*

*Download size depends on selected download components.

Overview

The Azure Services Platform is designed to help developers quickly and easily create, deploy, manage, and distribute web services and applications on the Internet. Windows Azure is an operating system for the cloud that serves as the development, run-time, and control environment for the Azure Services Platform. Windows Azure provides developers on-demand compute & storage to host, scale, and manage web applications on the internet through Microsoft data centers.

Windows Azure is elastic, flexible, and interoperable. With Windows Azure developers can achieve high levels of service availability

NOTE *There have been several different releases of the SDK. Check the Microsoft web site for the most current version.*

Microsoft Windows Azure

- The Google App Engine, which has been written for Windows, Linux, and Mac OS X, Azure only operates on the latest Windows machines.

- Take a look at the system requirements to see if your machine is capable of running the SDK.

Supported operating systems include :

- Windows Server 2008 Enterprise
- Windows Server 2008 Standard
- Windows Vista Business
- Windows Vista Business 64-bit edition
- Windows Vista Home Premium
- Windows Vista Home Premium 64-bit edition
- Windows Vista Ultimate
- Windows Vista Ultimate 64-bit edition

Microsoft Windows Azure

- If you have a 64-bit machine, you need to use the 64-bit version of the SDK. The 32-bit version will not work.

Other software requirements include :

- Windows Vista SP1 (when installing on Windows Vista)
- .NET Framework 3.5 SP1
- IIS 7.0 (with ASP.NET and WCF HTTP Activation)
- Microsoft SQL Server Express 2005 or Microsoft SQL Server Express 2008
- Windows PowerShell (optional)
- Microsoft also recommends that you have the Windows Azure Tools for Microsoft
- Visual Studio installed.
- Once you install the Azure SDK, it will add new templates:
- Blank Cloud Service
- Web Cloud Service
- Web and Worker Cloud Service
- Worker Cloud Service
- Windows Azure SDK includes two development utilities. This helps us develop our
- Windows Azure application in our local machine:

Microsoft Windows Azure

- The Development Fabric This tool helps us to run and test Windows Azure application locally before deploying into Cloud.
- The Development Storage This simulates the Blob, Queue, and Table Storage services available in the cloud. Enabling IIS 7.0 on Windows Server 2008.
- You will need to enable IIS 7.0 with WCF HTTP Activation on Windows Server 2008. To do this, follow these steps:

1. Click the Start button and then point to All Programs.
2. Point to Administrative Tools and then click Server Manager.
3. In Server Manager under Features Summary, click Add Features.
4. In the resulting dialog box, under .NET Framework 3.0, click .NET Framework 3.0.
5. Under WCF Activation, click HTTP Activation.

6. Click Next to start HTTP Activation.

7. In Server Manager, under Roles Summary, verify that Web Server (IIS) is on the list of available roles. If it is not on the list, click Add Roles to install Internet Information Services.

8. In Server Manager, under Roles Summary, click Web Server (IIS).

9. In the resulting window, click Add Role Services.

In the next window, expand Web Server, expand Common HTTP Features, and then click Static Content.

11. In the Add Role Services dialog box, expand Web Server, expand Application Development, and then click ASP.NET.

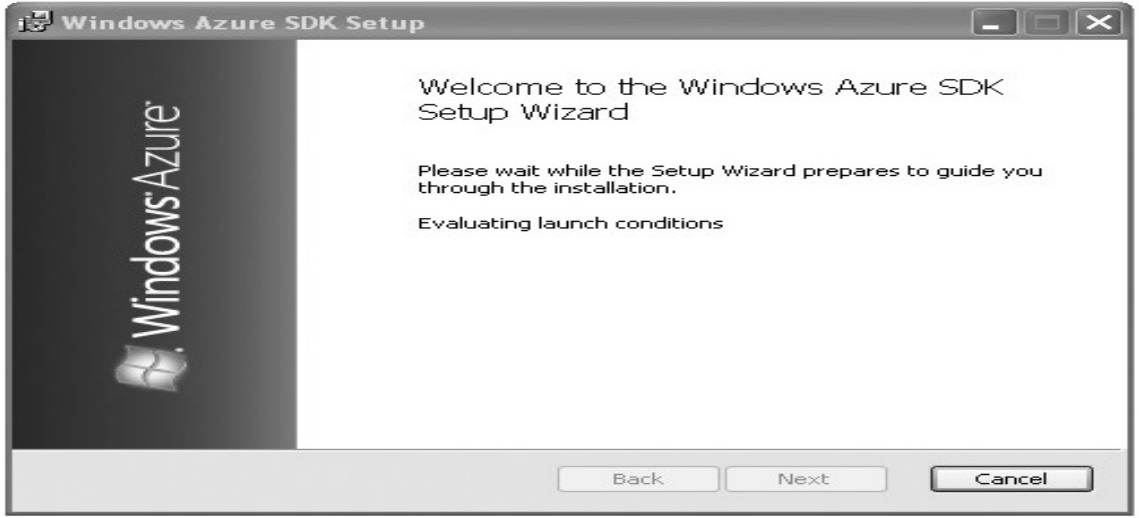
12. Click Next to enable Static Content and ASP.NET.

Microsoft Windows Azure

- Enabling IIS 7.0 on Windows Vista
- To enable IIS 7.0 with ASP.NET and WCF HTTP Activation on Windows Vista, follow these steps:
- 1. Click the Start button, click Settings, click Control Panel, click Programs, and then click Programs and Features.
- 2. Click Turn Windows Features On or Off.
- 3. Under Microsoft .NET Framework 3.0, click Windows Communication Foundation HTTP Activation.
- 4. Under Internet Information Services, expand World Wide Web Services.
- 5. Under Application Development Features, click ASP.NET.
- 6. Under Common HTTP Features, click Static Content.
- 7. Install the selected features.

Install the Windows Azure SDK :

- If you have an old version of the Windows Azure SDK on your machine, you must remove it before installing the new version. Right-click the Microsoft Windows Installer file, and then click Uninstall to remove the previous version of the SDK.
- Next, install the SDK on your computer by running the Windows Installer file.
- By default, the SDK is installed into the C:\Program Files\Windows Azure SDK\ directory.



Test the SDK :

- The SDK comes with a number of samples that you may find useful. Here, we're going to use them to test out the new installation.

- Navigate to the aforementioned SDK directory and unzip the file to a directory where you have write access.

To test the SDK installation, follow these steps:

1. Open the Windows Azure SDK command prompt by clicking the Start button, clicking Program Files, and then clicking Window Azure SDK (January 2009 CTP).
2. Navigate to the sample directory, and then run the RunDevStore.cmd utility to build the samples, create local tables required by the samples, and launch development storage. Running development storage starts the local Blob, Queue, and Table services.
3. Navigate to the HelloWorld application directory, and then run runme.cmd.
4. The development fabric icon will appear in the system tray after a moment.

Running the sample automatically launches your web browser and points to the service's default web page.
The web interface for the Hello World sample will be displayed in the browser window.

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4. The development fabric icon will appear in the system tray after a moment.
5. Running the sample automatically launches your web browser and points to the service's default web page.
6. The web interface for the Hello World sample will be displayed in the browser window.

Creating the App :

Creating cloud apps with Windows Azure is accomplished through Microsoft Visual Studio 2008. To start making your apps, follow these steps:

1. Run Visual Studio 2008 as an administrator. Click the Start button and then point to all programs.
2. Point to Microsoft Visual Studio 2008, right-click Microsoft Visual Studio 2008, and click Run as Administrator.
3. When the User Account Control box appears, click Continue.
4. From the File menu, click New, and then click Project.
5. In the resulting dialog box, select Cloud Services from the Visual C# project type.
6. In the Templates list, click Web Cloud Service. This creates a web role.
7. Enter the names for your project and solution, and make sure that the Create Directory for Solution box is checked.
8. In the Solution Explorer, you'll see that the solution structure was created. You'll see two projects, a Cloud Compute project and an ASP.NET project. The ASP.NET project will contain your code. The Cloud Compute project contains a reference to the ASP.NET project, as well as the service definition file (.csdef) and service configuration file (.cscfg).
9. Change the Default.aspx page to display some custom text for this example.

Running the App Locally :

Next, test the application locally before publishing it to the Azure cloud. You can do that by following these steps:

1. Press F5. This starts the service in the development fabric.

2. Expand the tree on the left panel to see the service deployments.

3. Expand the HelloFabric node to see the web role and worker role that are running within the service.

4. Expand the node for the web role or worker role to view the running instances of the role.

5. Click on a numbered node to see the messages being written to the log for this role instance.

6. Click the Service Details node. This shows you the roles and ports that were requested and issued.

7. Switch back to Visual Studio, and press SHIFT+F5 to stop debugging

Troubleshooting :

- Troubleshooting on the cloud is a different animal than in a traditional IT environment.
- Conventional troubleshooting tools and processes were developed around the hub-and-spoke concept, with remote applications being the spokes to the centralized datacenter being the hub.
- Applications delivered from an off-site provider's datacenter defy traditional monitoring and require a different way to troubleshoot.
- At your organization, you have local visibility and control of applications. That control is much more limited with SaaS applications. In particular, it is difficult to distinguish SaaS traffic from other Internet usage going in and coming out of remote locations.
- IT staff do not have the local ability to determine whether the service is running or if the client has connectivity to the hosted application.
- Deep packet inspection (DPI) can deliver this granularity, but this technology is expensive to deploy throughout the enterprise.

Microsoft Windows Azure

- Another issue is that when users face performance problems or have other issues, they are not going to contact the SaaS provider. They're going to call the IT support desk.
- Before your IT department relays the problem to the SaaS provider, they must first pursue the problem on their end to rule out local problems.
- For instance, there could be a legitimate issue with Amazon, for which you need to invoke the service level agreement (SLA). On the other hand, the problem could simply be the result of contention for the Internet.
- The problem could be exacerbated by SaaS users within the enterprise working from different remote sites, using different types of access and competing for resources. This can quickly eat up bandwidth.
- The user could be on a wireless LAN competing for shared bandwidth with other applications that have fluctuating usage needs, are experiencing slowdowns, or have issues with signal interference.

Microsoft Windows Azure

- To troubleshoot SaaS problems, you must be able to understand the perspective of the application and the end user, and to see all the variations inside and outside the infrastructure.
- Visibility is not available with traditional management tools and technology, but new tools are available.
- By using this tracking information, profiles are created for all applications and for the networked application experiences of each end user that allow IT to detect unusual behavior and figure out the root cause of the performance problems.
- IT can discover hidden causes of SaaS performance problems by figuring out which applications are generating high bandwidth on the link or by finding applications that are causing congestion.
- When IT staff has this information, they can quickly shut down problematic users and alert SaaS providers about problems at their datacenter.

Application Management

- Once you've got your application on the cloud, you need to be able to manage it. While the application isn't in your datacenter, there are still ways you can manage it.
- A lot will depend on the terms of service you have with your cloud provider. They may not allow for you to manage the applications at all.
- When you do decide to manage your cloud application, you can use a product like Kaavo's (www.kaavo.com) cloud application management software: **Infrastructure and Middleware on Demand (IMOD).**
- IMOD is the first solution with an application-focused approach to IT infrastructure management through public and private clouds.
- Companies traditionally manage their servers individually, which is complex and costly and impedes business growth across an enterprise.

Application Management

- By tapping the capabilities of cloud computing, however, IMOD enables users to manage infrastructure as a unified system and **provides the following benefits:**
- Application and service–centric n-tier configuration IMOD automatically brings online one or multiserver systems for running applications.
- Business continuity An interface to schedule automatic data backups ensures business continuity.
- Security and access control IMOD provides a point-and-click interface to secure data in the clouds through the National Security Association's recommended AES 256-bit data encryption.
- It allows users to easily and securely connect to servers, transfer data to and from internal datacenters, and configure custom firewall rules on cloud servers.
- Effective monitoring and alerts Users can monitor resources used by their applications and set up alerts to proactively manage application service levels.

Application Management

- "Moving infrastructure to the cloud is gaining momentum. Kaavo will enable organizations to effectively reap the benefits of cloud computing and to maximize fewer resources.
- This is particularly critical given current economic conditions and the growing need for companies to reduce their carbon footprint," said Jamal Mazhar, founder and CEO of Kaavo.
- "Through patent-pending technology, we are pleased to launch IMOD and provide simplicity, flexibility, and security for users. This in turn, allows them to focus on innovation and core business activities necessary for business growth."
- IMOD is available for a free 30-day trial at https://imod.kaavo.com. Developing your own apps is definitely doable, and there is no lack of places that would welcome you to host your apps on their clouds.
- The differences come down to the features they offer and how much they charge, but if there is a specific function you want done, just point your programmers to the cloud and get them started.